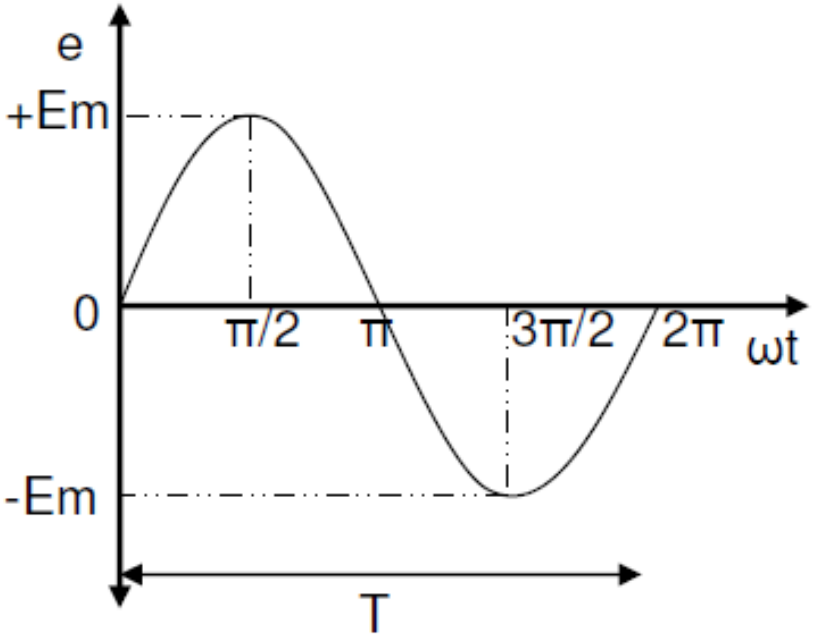
UNIT-II

Single Phase AC circuits

Definition of Alternating Quantity



$v(t) = Vm \sin\theta$ or ωt

$i(t) = Im \sin\theta$ or ωt

Where Vm, Im are maximum or peak values of voltage and current
ω is angular frequency in rad/sec

Alternating current is the current which constantly changes in amplitude, and which reverses direction at regular intervals.

1. **Amplitude:** It is the maximum value attained by an alternating quantity. Also called as maximum or peak value.

2. **Time Period (T):** It is the Time Taken in seconds to complete one cycle of an alternating quantity.

3. **Instantaneous Value:** It is the value of the quantity at any instant.

4. Waveform: The shape of the curve, which connects all the instantaneous values is known as waveform.

5. Frequency (f): It is the number of cycles that occur in one second. The unit for frequency is Hz or cycles/sec. The relationship between frequency and time period can be derived as follows.

$$T = 1/f$$

- **Angular Frequency (ω):** Angular frequency is defined as the number of radians covered in one second(i.e., the angle covered by the rotating coil). The unit of angular frequency is rad/sec.

$$\omega = \frac{2\pi}{T} = 2\pi f$$

- **Problem 1:**

An alternating current i is given by **i = 141.4 sin 314t.** Find i) The maximum value ii) Frequency iii) Time Period iv) The instantaneous value when t=3ms

$i = 141.4 \sin 314t$

i) Maximum value Im=141.4 A

$\omega = 314$ rad/sec

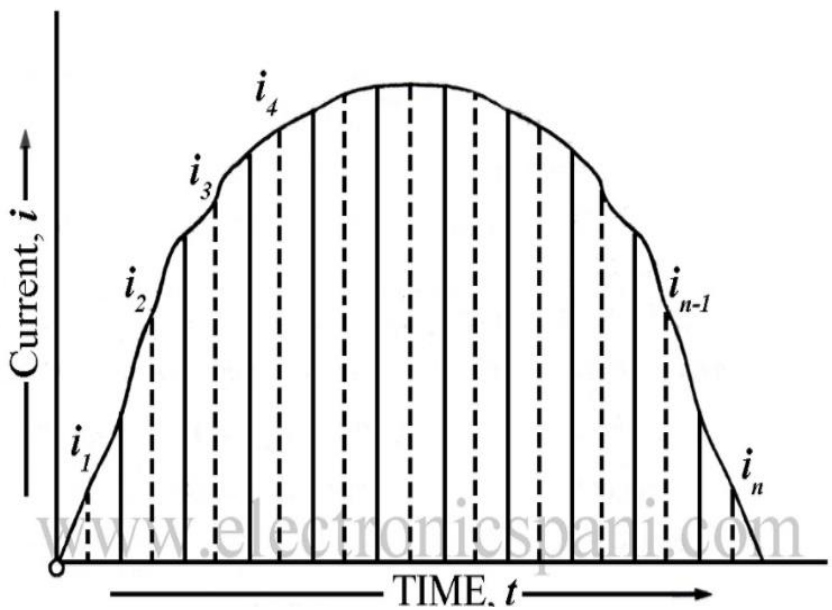
ii) $f = \omega / 2\pi = 50$ Hz As $\omega = 2\pi f$

iii) $T = 1/f = 0.02$ sec

iv) $i = 141.4 \sin(314 \times 0.003) = 114.35$A

Average Value

- The arithmetic average of all the instantaneous values of an alternating quantity over one cycle is called its average value



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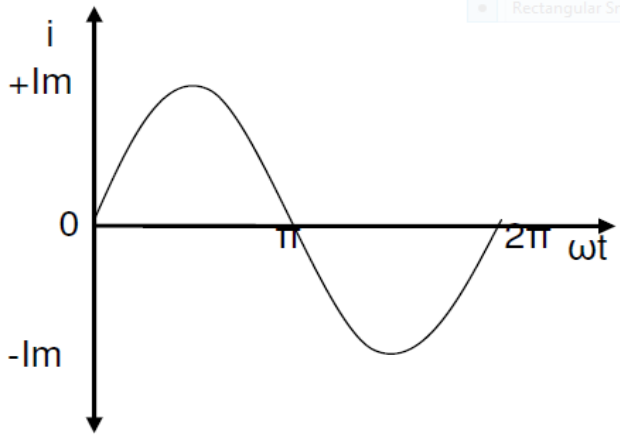
$$I_{av} = \frac{i_1 + i_2 + i_3 + i_4 + i_n}{n}$$

• Average value of a sinusoidal current:



For symmetrical Waveforms: Average value = Area under one half cycle/ Base

$$V_{av} = \frac{1}{\pi}\int_0^{\pi} v\, d(\omega t)$$

For Asymmetrical Waveforms: Average value = Area under one cycle/ Base

$$V_{av} = \frac{1}{2\pi}\int_0^{2\pi} v\, d(\omega t)$$

Derivation of Average value:

$$I_{avg} = \frac{1}{\pi}\int_0^{\pi} i(t).\, d(\omega t)$$

$$= \frac{1}{\pi}\int_0^{\pi} I_m \sin\omega t.\, d(\omega t)$$

$$= \frac{I_m}{\pi}\left[-\cos\omega t\right]$$

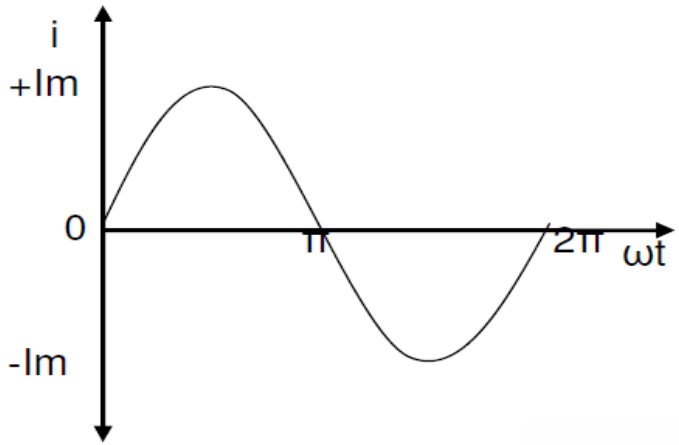
$$= \frac{I_m}{\pi}\left[1+1\right] = \frac{2I_m}{\pi}$$

$$\boxed{I_{avg} = \frac{2I_m}{\pi} = 0.637\, I_m}$$

(or)

$$\boxed{V_{avg} = \frac{2V_m}{\pi} = 0.637\, V_m}$$

• RMS value of a sinusoidal current:



$$RMS = \sqrt{\frac{\text{Area under squared curve}}{base}}$$

For symmetrical Waveforms:

$$\text{RMS value} = \sqrt{\frac{1}{\pi}\int_0^{\pi} v^2(\omega t).\, d(\omega t)}$$

For Asymmetrical Waveforms:

$$V_{rms} = \sqrt{\frac{1}{2\pi}\int_0^{2\pi} v^2 d(\omega t)}$$

Derivation of RMS value:

$$\text{RMS value} = \sqrt{\frac{1}{\pi}\int_0^{\pi} v^2(\omega t)\cdot d(\omega t)}$$

$$= \left\{\frac{1}{\pi}\left[\int_0^{\pi} V_m^2\cdot \sin^2\omega t\cdot d(\omega t)\right]\right\}^{1/2}$$

$$= \left\{\frac{V_m^2}{\pi}\left[\int_0^{\pi}\left(\theta/2 - \frac{\cos 2\omega t}{2}\right)\right]\right\}^{1/2}$$

$$= \left\{\frac{V_m^2}{\pi}\left(\tfrac{1}{2}(\pi - 0) - \left(\frac{\sin 2\pi}{4} - \frac{\sin 0}{4}\right)\right)\right\}^{1/2}$$

$$= \frac{V_m}{\sqrt{\pi}}\left(\pi/2 - 0\right)^{1/2}$$

$$= \frac{V_m}{\cancel{\sqrt{\pi}}} \times \frac{\cancel{\sqrt{\pi}}}{\sqrt{2}} = \frac{V_m}{\sqrt{2}}$$

$$\therefore \boxed{V_{RMS} = \frac{V_m}{\sqrt{2}}} \quad \text{\&}^{\text{lly}} \quad \boxed{I_{RMS} = \frac{I_m}{\sqrt{2}}}$$

Form Factor:

- The ratio of RMS value to the average value of an alternating quantity is known as Form Factor

$$FF = \frac{\text{RMS VALUE}}{\text{AVERAGE VALUE}}$$

Peak Factor or Crest Factor:

- The ratio of maximum value to the RMS value of an alternating quantity is known as the peak factor

$$PF = \frac{\text{MAXIMUM VALUE}}{\text{RMS VALUE}}$$

For a sinusoidal waveform:

$$I_{av} = \frac{2I_m}{\pi} = 0.637 I_m$$

$$I_{rms} = \frac{I_m}{\sqrt{2}} = 0.707 I_m$$

$$FF = \frac{I_{rms}}{I_{av}} = \frac{0.707 I_m}{0.637 I_m} = 1.11$$

$$PF = \frac{I_m}{I_{rms}} = \frac{I_m}{0.707 I_m} = 1.414$$

Find the average value and rms value of the waveform shown in Fig.

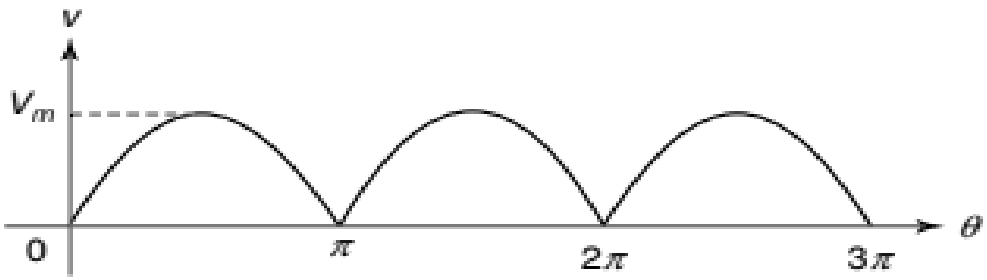


Fig.

$$v = V_m \sin\theta \qquad 0 < \theta < \pi$$

Solution

(a) Average value of the waveform

$$V_{avg} = \frac{1}{\pi}\int_0^{\pi} v(\theta)\, d\theta = \frac{1}{\pi}\int_0^{\pi} V_m \sin\theta\, d\theta = \frac{V_m}{\pi}\left[-\cos\theta\right]_0^{\pi}$$

$$= \frac{V_m}{\pi}[1+1] = 0.637\, V_m$$

(b) rms value of the waveform

$$V_{rms} = \sqrt{\frac{1}{\pi}\int_0^{\pi} v^2(\theta)\, d\theta} = \sqrt{\frac{1}{\pi}\int_0^{\pi} V_m^2 \sin^2\theta\, d\theta} = \sqrt{\frac{V_m^2}{\pi}\int_0^{\pi} \sin^2\theta\, d\theta}$$

$$= \sqrt{\frac{V_m^2}{\pi}\int_0^{\pi}\left(\frac{1-\cos 2\theta}{2}\right)d\theta} = \sqrt{\frac{V_m^2}{\pi}\left[\frac{\theta}{2}-\frac{\sin 2\theta}{4}\right]_0^{\pi}} = \sqrt{\frac{V_m^2}{\pi}\left[\frac{\pi}{2}-\frac{\sin 2\pi}{4}-0+\frac{\sin 0}{4}\right]} = 0.707\,V_m$$

Find the average and rms values of the waveform shown in Fig

Fig. ·

$$v = V_m \sin\theta \qquad 0 < \theta < \pi$$
$$= 0 \qquad \pi < \theta < 2\pi$$

Solution

(a) Average value of the waveform

$$V_{\text{avg}} = \frac{1}{2\pi}\int_0^{2\pi} v(\theta)\, d\theta = \frac{1}{2\pi}\left[\int_0^{\pi} V_m \sin\theta\, d\theta + \int_{\pi}^{2\pi} 0\, d\theta\right]$$

$$= \frac{1}{2\pi}\int_0^{\pi} V_m \sin\theta\, d\theta = \frac{V_m}{2\pi}\left[-\cos\theta\right]_0^{\pi} = \frac{V_m}{2\pi}[1+1] = 0.318\, V_m$$

(b) rms value of the waveform

$$V_{\text{rms}} = \sqrt{\frac{1}{2\pi}\int_0^{2\pi} v^2(\theta)\, d\theta} = \sqrt{\frac{1}{2\pi}\left[\int_0^{\pi} V_m^2 \sin^2\theta\, d\theta + \int_{\pi}^{2\pi} 0\, d\theta\right]}$$

$$= \sqrt{\frac{1}{2\pi}\int_0^{\pi} V_m^2 \sin^2\theta\, d\theta} = \sqrt{\frac{V_m^2}{2\pi}\int_0^{\pi} \sin^2\theta\, d\theta} = \sqrt{\frac{V_m^2}{2\pi}\int_0^{\pi}\left(\frac{1-\cos 2\theta}{2}\right) d\theta}$$

$$= \sqrt{\frac{V_m^2}{2\pi}\left[\frac{\theta}{2} - \frac{\sin 2\theta}{4}\right]_0^{\pi}} = \sqrt{\frac{V_m^2}{2\pi}\left[\frac{\pi}{2} - \frac{\sin 2\pi}{4} - 0 + \frac{\sin 0}{4}\right]} = 0.5\ V_m$$

Find the average value and rms value of the waveform shown in Fig.

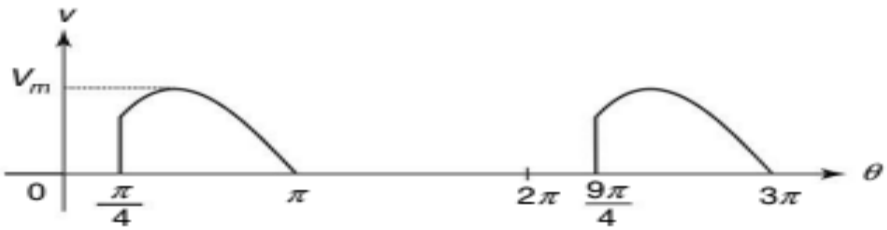


Fig.

$$v = 0 \qquad 0 < \theta < \frac{\pi}{4}$$

$$= V_m \sin\theta \qquad \frac{\pi}{4} < \theta < \pi$$

$$= 0 \qquad \pi < \theta < 2\pi$$

Solution

(a) Average value of the waveform

$$V_{avg} = \frac{1}{2\pi}\int_0^{2\pi} v(\theta)\, d\theta = \frac{1}{2\pi}\int_{\frac{\pi}{4}}^{\pi} V_m \sin\theta\, d\theta$$

$$= \frac{V_m}{2\pi}\left[-\cos\theta\right]_{\frac{\pi}{4}}^{\pi} = \frac{V_m}{2\pi}[1+0.707] = 0.272 \text{ Vm}$$

(b) rms value of the waveform

$$V_{rms} = \sqrt{\frac{1}{2\pi}\int_0^{2\pi} v^2(\theta)\, d\theta} = \sqrt{\frac{1}{2\pi}\int_{\frac{\pi}{4}}^{\pi} V_m^2 \sin^2\theta\, d\theta} = \sqrt{\frac{V_m^2}{2\pi}\int_{\frac{\pi}{4}}^{\pi} \sin^2\theta\, d\theta} = \sqrt{\frac{V_m^2}{2\pi}\int_{\frac{\pi}{4}}^{\pi}\left(\frac{1-\cos 2\theta}{2}\right) d\theta}$$

$$= \sqrt{\frac{V_m^2}{2\pi}\left[\frac{\theta}{2} - \frac{\sin 2\theta}{4}\right]_{\frac{\pi}{4}}^{\pi}} = \sqrt{\frac{V_m^2}{2\pi}\left[\frac{\pi}{2} - \frac{\sin 2\pi}{4} - \frac{\pi}{8} + \frac{\sin\frac{\pi}{2}}{4}\right]} = 0.476\, V_m$$

Find the average and rms values of the waveform shown in Fig.

$$v = V_m \qquad 0 < t < \frac{T}{2}$$

$$= 0 \qquad \frac{T}{2} < t < T$$

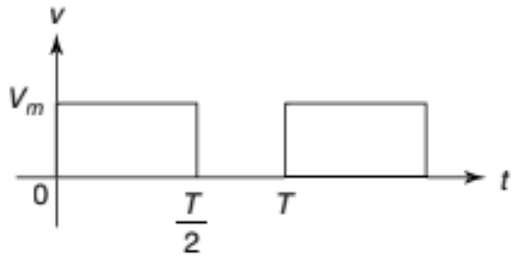


Fig.

(a) Average value of the waveform

$$V_{avg} = \frac{1}{T}\int_0^T v(t)\,dt = \frac{1}{T}\left[\int_0^{\frac{T}{2}} V_m\,dt + \int_{\frac{T}{2}}^{T} 0\,dt\right] = \frac{1}{T}\int_0^{\frac{T}{2}} V_m\,dt$$

$$= \frac{V_m}{T}[t]_0^{\frac{T}{2}} = \frac{V_m}{T}\cdot\frac{T}{2} = 0.5\,V_m$$

(b) rms value of the waveform

$$V_{rms} = \sqrt{\frac{1}{T}\int_0^T v^2(t)\,dt} = \sqrt{\frac{1}{T}\int_0^{\frac{T}{2}} V_m^2\,dt} = \sqrt{\frac{V_m^2}{T}[t]_0^{\frac{T}{2}}}$$

$$= \sqrt{\frac{V_m^2}{T}\cdot\frac{T}{2}} = 0.707\,V_m$$

Find the average and rms values of the waveform shown in Fig.

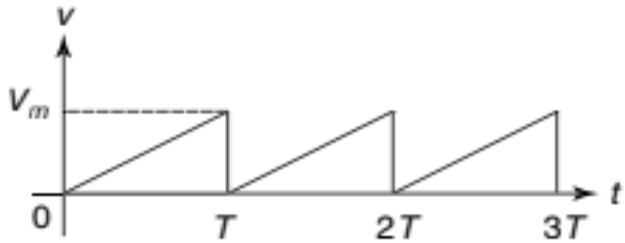


Fig.

$$v = \frac{V_m}{T} t \qquad 0 < t < T$$

Ans: Vavg= 0.5Vm and Vrms= 0.577Vm

A full-wave rectified wave is clipped at 70.7% of its maximum value as shown in Fig. Find its average and rms values.

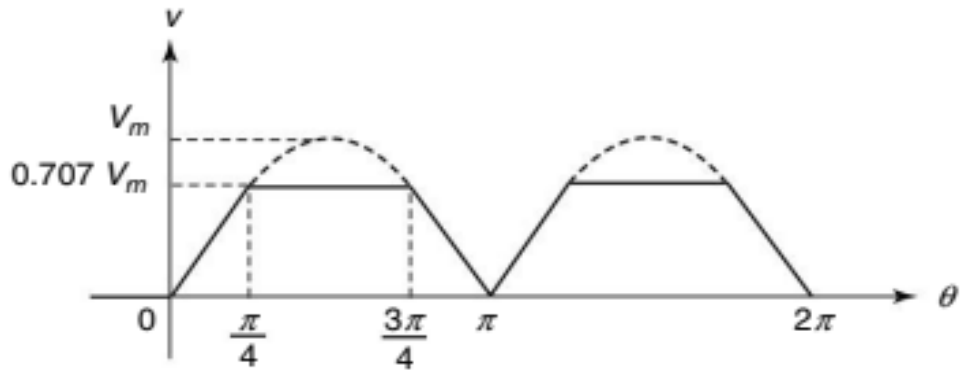


Fig.

$$v = V_m \sin\theta \qquad 0 < \theta < \frac{\pi}{4}$$

$$= 0.707\, V_m \qquad \frac{\pi}{4} < \theta < \frac{3\pi}{4}$$

$$= V_m \sin\theta \qquad \frac{3\pi}{4} < \theta < \pi$$

Ans: Vavg= 0.54Vm and Vrms= 0.584Vm

Find the average and rms values of the waveform shown in Fig.

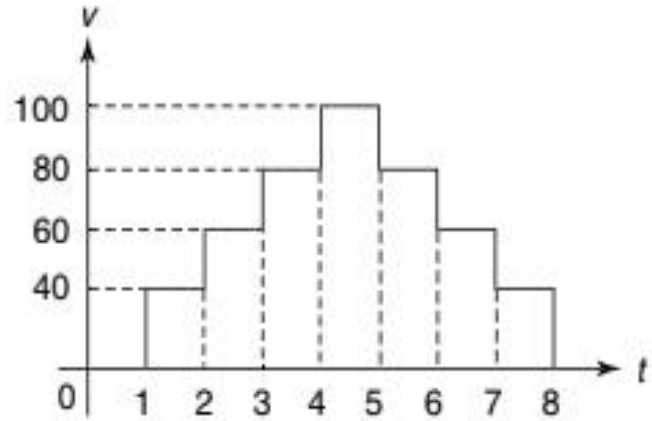


Fig.

Solution

(a) Average value of the waveform

$$V_{avg} = \frac{0+40+60+80+100+80+60+40}{8} = 57.5 \text{ V}$$

(b) rms value of the waveform

$$V_{rms} = \sqrt{\frac{0^2+(40)^2+(60)^2+(80)^2+(100)^2+(80)^2+(60)^2+(40)^2}{8}} = 64.42 \text{ V}$$

Find the average and rms values of the waveform shown in Fig.

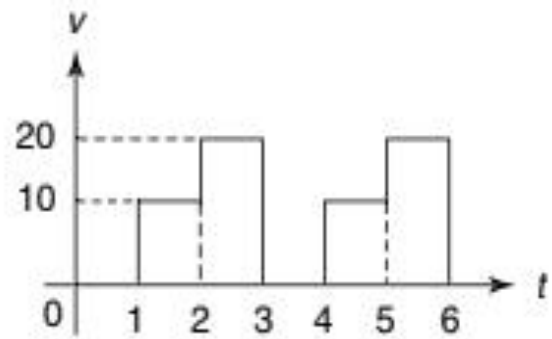


Fig.

Solution

(a) Average value of the waveform

$$V_{avg} = \frac{0+10+20}{3} = 10 \text{ V}$$

(b) rms value of the waveform

$$V_{rms} = \sqrt{\frac{0^2+(10)^2+(20)^2}{3}} = 12.9 \text{ V}$$

Find the effective value of a resultant current in a wire which carries simultaneously a direct current of $i_1 = 10\ A$ and alternating current given by $i_2 = 12 \sin \omega t + 6 \sin\left(3\omega t - \frac{\pi}{6}\right) + 4 \sin\left(5\omega t + \frac{\pi}{3}\right)$.

Solution $i_1 = 10$ A

$$i_2 = 12 \sin \omega t + 6 \sin\left(3\omega t - \frac{\pi}{6}\right) + 4 \sin\left(5\omega t + \frac{\pi}{3}\right)$$

$$i = i_1 + i_2 = 10 + 12 \sin \omega t + 6 \sin\left(3\omega t - \frac{\pi}{6}\right) + 4 \sin\left(5\omega t + \frac{\pi}{3}\right)$$

$$= 10 + 12 \sin\theta + 6 \sin(3\theta - 30°) + 4 \sin(5\theta + 60°)$$

$$I_{eff} = \sqrt{\frac{1}{2\pi}\int_0^{2\pi} i^2(\theta)\, d\theta} = \sqrt{\frac{1}{2\pi}\int_0^{2\pi} \left[10 + 12 \sin\theta + 6\sin(3\theta - 30°) + 4\sin(5\theta + 60°)\right]^2 d\theta} = 14.07 \text{ A}$$

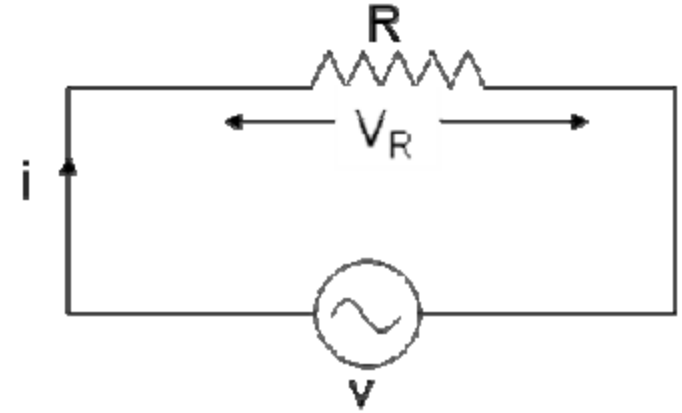
AC circuit with a pure Resistor:

$$v = V_m \sin \omega t \qquad \text{---------- (1)}$$

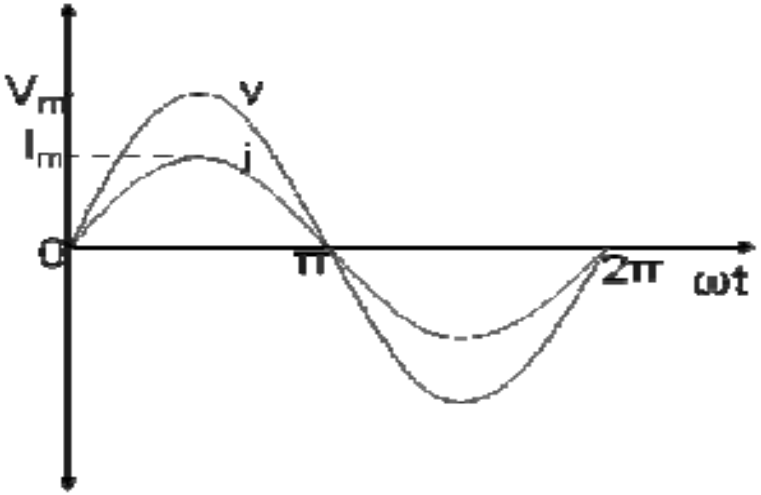
$$i = \frac{v}{R} = \frac{V_m \sin \omega t}{R}$$

$$i = I_m \sin \omega t \qquad \text{----------(2)}$$

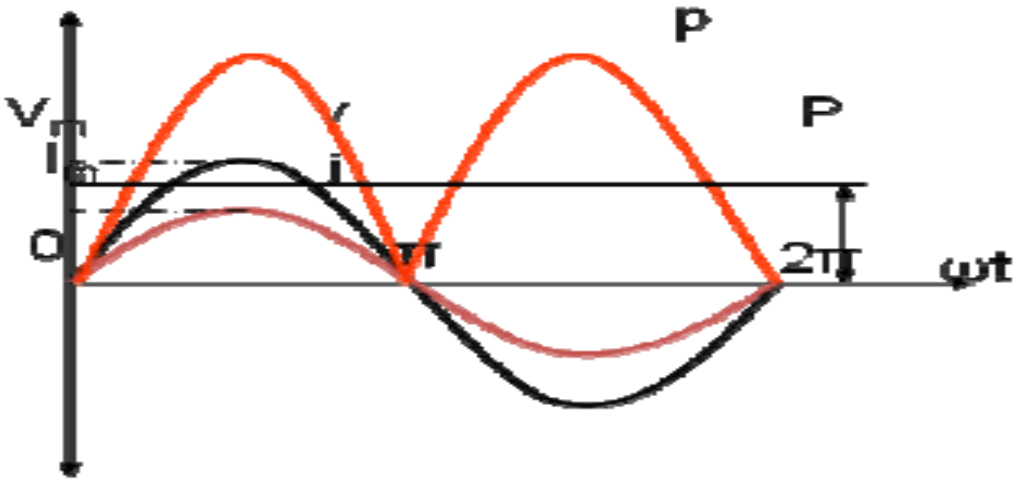
Where $I_m = \dfrac{V_m}{R}$



From equation (1) and (2) we conclude that in a pure resistive circuit, the **voltage and current are in phase**.



V in phase with I



Instantaneous power:

$$p = vi$$

$$p = (V_m \sin \omega t)(I_m \sin \omega t)$$

$$p = V_m I_m \sin^2 \omega t$$

$$p = \frac{V_m I_m}{2}(1 - \cos 2\omega t)$$

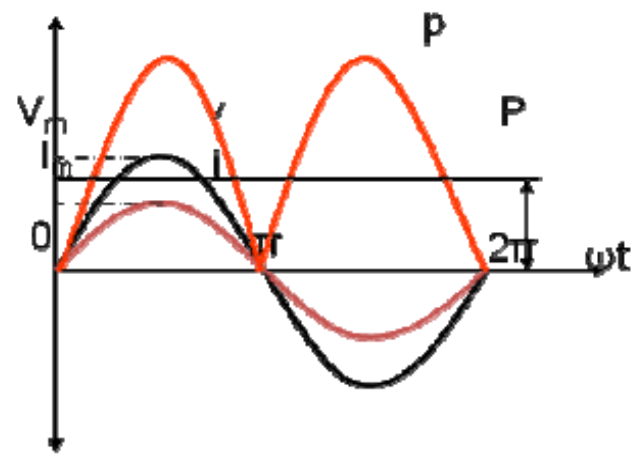
$$p = \frac{V_m I_m}{2} - \frac{V_m I_m}{2} \cos 2\omega t$$

- The instantaneous power consists of two terms. The first term is called as the constant power term and the second term is called as the fluctuating power term.
- **Average power:** From the instantaneous power we can find the average power over one cycle as follows

$$P = \frac{1}{2\pi}\int_0^{2\pi}\left[\frac{V_m I_m}{2} - \frac{V_m I_m}{2}\cos 2\omega t\right] d\omega t$$

$$P = \frac{V_m I_m}{2} = \frac{V_m}{\sqrt{2}}\frac{I_m}{\sqrt{2}}$$

$$P = V.I$$



The average power is the product of the rms voltage and the rms current.

Problems

1. An ac circuit consists of a pure resistance of 10Ω and is connected to an ac supply of 230 V, 50 Hz. Calculate the (i) current (ii) power consumed and (iii) equations for voltage and current.

$$(i) I = \frac{V}{R} = \frac{230}{10} = 23A$$

$$(ii) P = VI = 230 \times 23 = 5260W$$

$$(iii) V_m = \sqrt{2}V = 325.27V$$

$$I_m = \sqrt{2}I = 32.52A$$

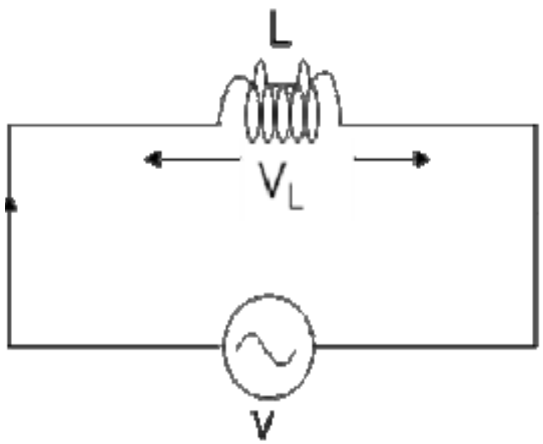
$$\omega = 2\pi f = 314 rad/\sec$$

$$v = 325.25 \sin 314t$$

$$i = 32.52 \sin 314t$$

AC circuit with a pure Inductor

$$v = V_m \sin \omega t \qquad \text{---------- (1)}$$



We can find the current through the inductor as follows

$$v = L\frac{di}{dt}$$

$$V_m \sin \omega t = L\frac{di}{dt}$$

$$di = \frac{V_m}{L}\sin \omega t dt$$

$$i = \frac{V_m}{L}\int \sin \omega t dt$$

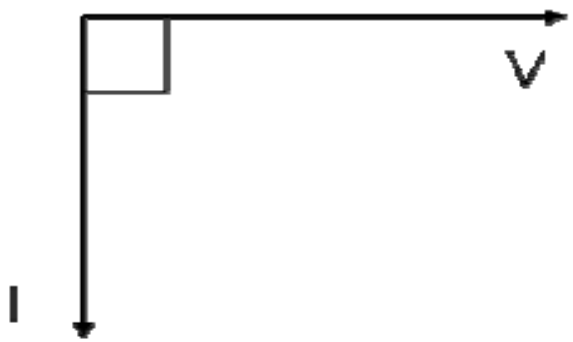
$$i = \frac{V_m}{\omega L}(-\cos \omega t)$$

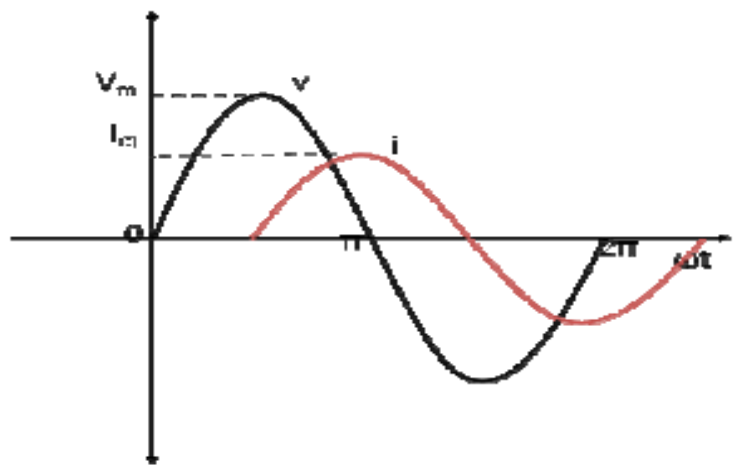
$$i = \frac{V_m}{\omega L}\sin(\omega t - \pi/2)$$

$$i = I_m \sin(\omega t - \pi/2) \qquad \text{----------(2)}$$

Where $\quad I_m = \dfrac{V_m}{\omega L}$

From equation (1) and (2) we observe that in a pure inductive circuit, **the current lags behind the voltage by 90º.**





The inductive reactance X_L is given as

$$X_L = \omega L = 2\pi f L$$

Instantaneous power:

$$p = vi$$

$$p = (V_m \sin \omega t)(I_m \sin(\omega t - \pi / 2))$$

$$p = -V_m I_m \sin \omega t \cos \omega t$$

$$p = -\frac{V_m I_m}{2} \sin 2\omega t$$

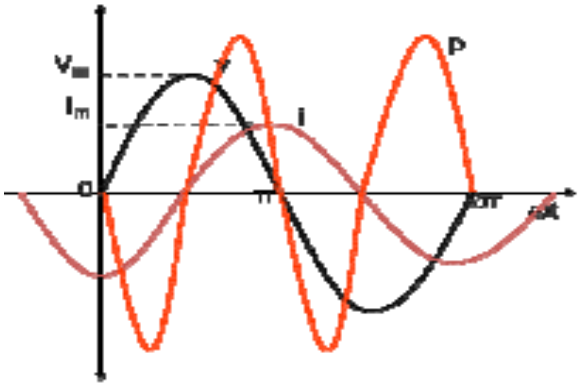
As seen from the above equation, the instantaneous power is fluctuating in nature.

Average power

From the instantaneous power we can find the average power over one cycle as follows

$$P = \frac{1}{2\pi} \int_0^{2\pi} -\frac{V_m I_m}{2} \sin 2\omega t d\omega t$$

$$P = 0$$



The average power in a pure inductive circuit is zero.

2. A pure inductive coil allows a current of 10A to flow from a 230V, 50 Hz supply. Find (i) inductance of the coil (ii) power absorbed and (iii) equations for voltage and current.

$$(i) X_L = \frac{V}{I} = \frac{230}{10} = 23\Omega$$

$$X_L = 2\pi f L$$

$$L = \frac{X_L}{2\pi f} = 0.073H$$

$$(ii) P = 0$$

$$(iii) V_m = \sqrt{2}V = 325.27V$$

$$I_m = \sqrt{2}I = 14.14A$$

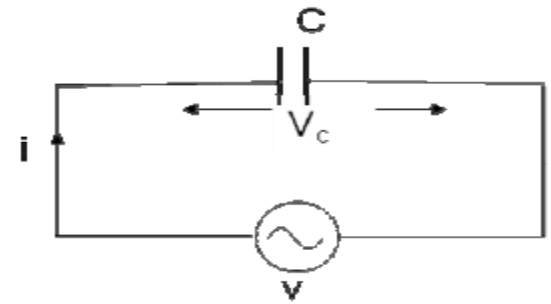
$$\omega = 2\pi f = 314 rad/\sec$$

$$v = 325.25 \sin 314t$$

$$i = 14.14 \sin(314t - \pi/2)$$

AC circuit with a pure Capacitor

$$v = V_m \sin \omega t \qquad \text{---------- (1)}$$



$$q = Cv$$

$$q = CV_m \sin \omega t$$

$$i = \frac{dq}{dt}$$

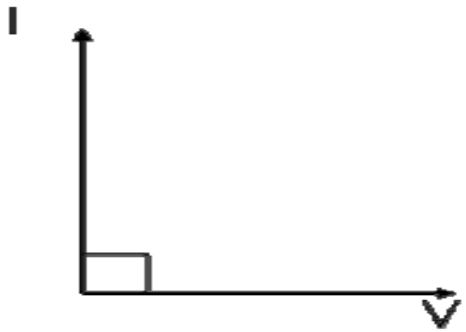
$$i = CV_m \omega \cos \omega t$$

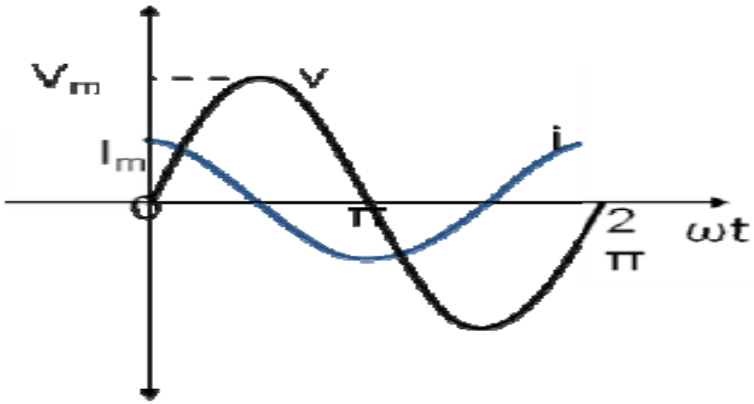
$$i = \omega CV_m \sin(\omega t + \pi / 2)$$

$$i = I_m \sin(\omega t + \pi / 2) \qquad \text{----------(2)}$$

Where $\quad I_m = \omega CV_m$

From equation (1) and (2) we observe that in a pure capacitive circuit, **the current leads the voltage by 90º.**





Capacitive reactance

The capacitive reactance X_C is given as

$$X_L = \frac{1}{\omega C} = \frac{1}{2\pi f C}$$

Instantaneous power

$$p = vi$$

$$p = (V_m \sin \omega t)(I_m \sin(\omega t + \pi/2))$$

$$p = V_m I_m \sin \omega t \cos \omega t$$

$$p = \frac{V_m I_m}{2} \sin 2\omega t$$

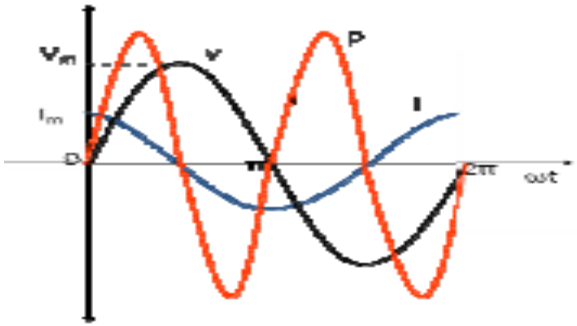
As seen from the above equation, the instantaneous power is fluctuating in nature.

Average power

From the instantaneous power we can find the average power over one cycle as follows

$$P = \frac{1}{2\pi} \int_0^{2\pi} \frac{V_m I_m}{2} \sin 2\omega t d\omega t$$

$$P = 0$$



The average power in a pure capacitive circuit is zero. Or in other words, the power consumed by a pure capacitance is zero.

3. A 318μF capacitor is connected across a 230V, 50 Hz system. Find (i) the capacitive reactance (ii) rms value of current and (iii) equations for voltage and current.

$$(i)\, X_C = \frac{1}{2\pi f C} = 10\Omega$$

$$(ii)\, I = \frac{V}{X_C} = 23A$$

$$(iii)\, V_m = \sqrt{2}V = 325.27V$$

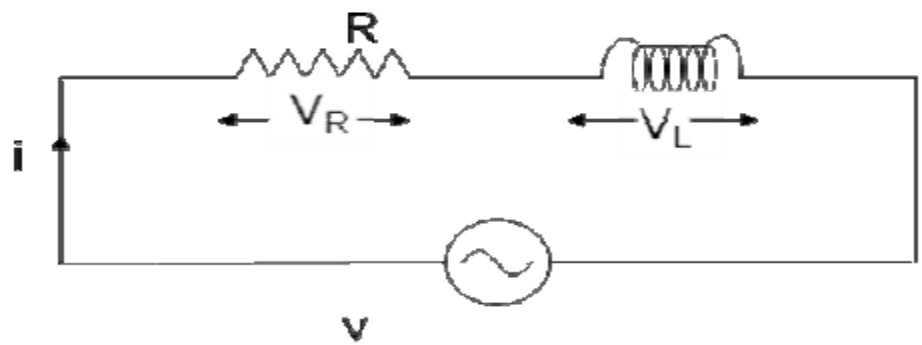
$$I_m = \sqrt{2}I = 32.53A$$

$$\omega = 2\pi f = 314 rad/\sec$$

$$v = 325.25 \sin 314t$$

$$i = 32.53 \sin(314t + \pi/2)$$

Series R-L circuit

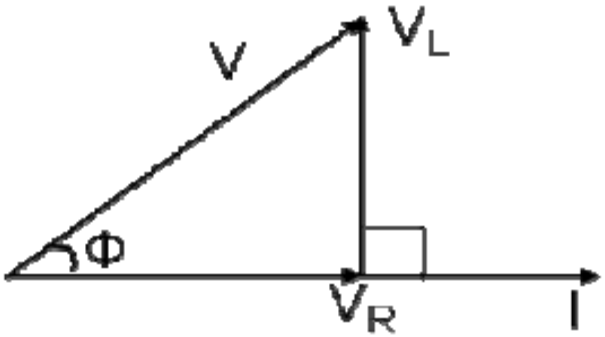


$$v = V_m \sin \omega t$$

The current flowing in the circuit is i. The voltage across the resistor is VR and that across the inductor is VL.

V_R=IR is in phase with I

V_L=IX_L leads current by 90 degrees



$$V = \sqrt{V_R^2 + V_L^2}$$

$$V_R = IR$$

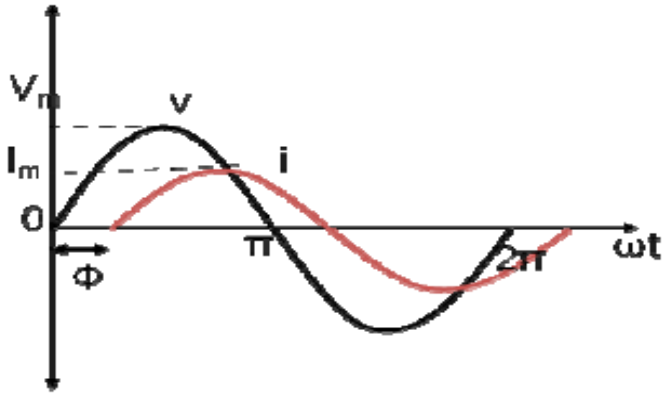
$$V_L = IX_L$$

$$V = \sqrt{(IR)^2 + (IX_L)^2}$$

$$V = I\sqrt{R^2 + X_L^2}$$

$$V = IZ$$

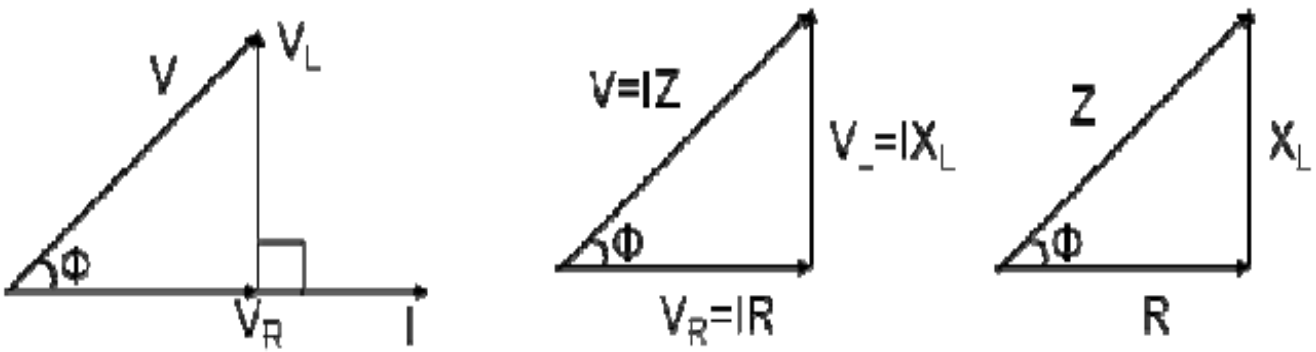
Where impedance $Z = \sqrt{R^2 + X_L^2}$



$$V = V_m \sin \omega t$$

$$I = I_m \sin(\omega t - \Phi)$$

Impedance triangle



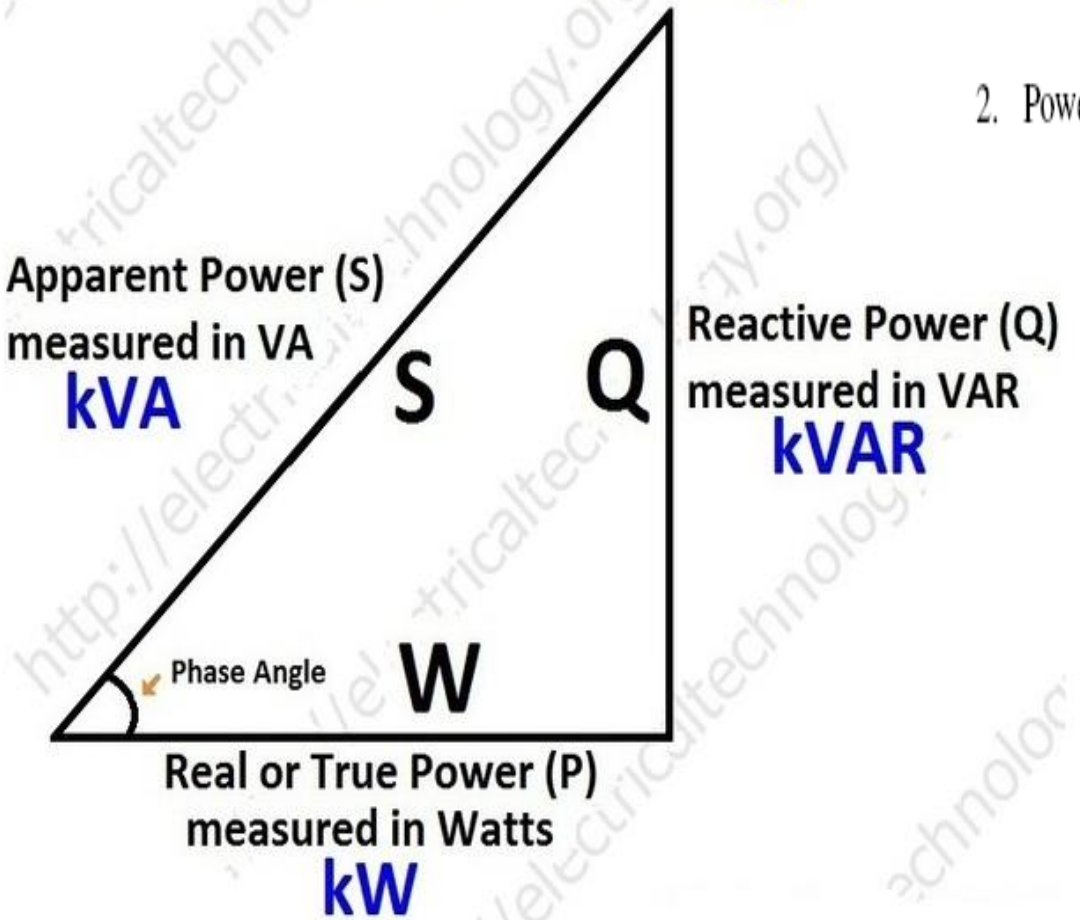
1. Impedance $Z = \sqrt{R^2 + X_L^2}$

2. Power Factor $\cos\Phi = \dfrac{R}{Z}$

3. Phase angle $\Phi = \tan^{-1}\left(\dfrac{X_L}{R}\right)$

4. Whether current leads or lags behind the voltage

Power Triangle

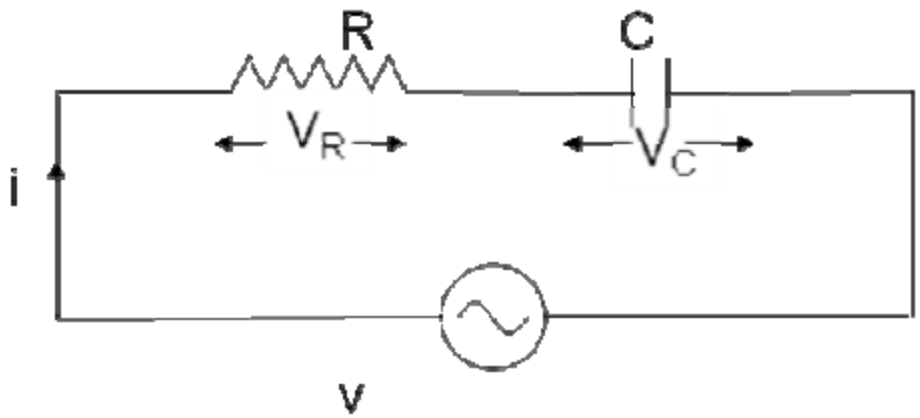


1. Apparent power $S = \sqrt{P^2 + Q^2}$

2. Power Factor $Cos\Phi = \frac{P}{S} = \frac{RealPower}{ApparentPower}$

P=V ICosΦ
Q=VISin Φ
S=VI

Series R-C circuit

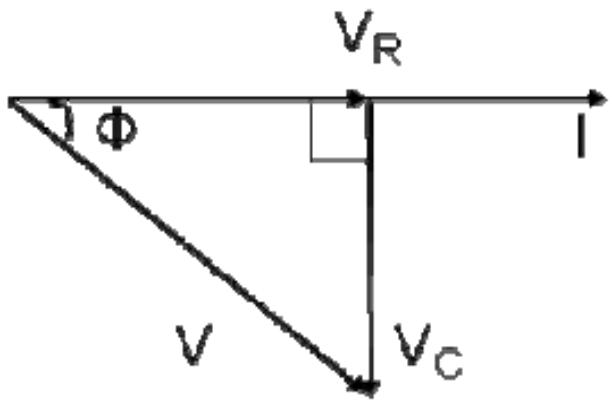


$$v = V_m \sin \omega t$$

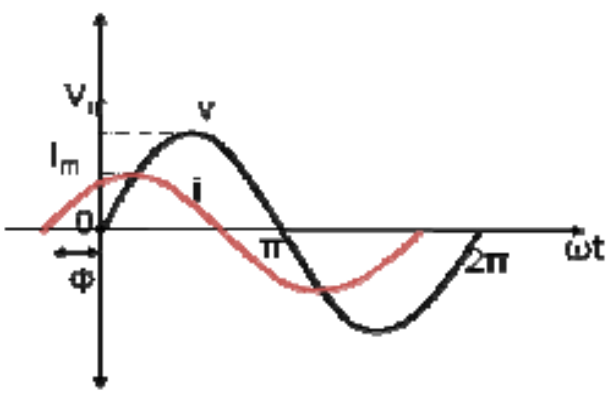
The current flowing in the circuit is i. The voltage across the resistor is VR and that across the capacitor is VC.

V_R=IR is in phase with I

V_C=IX_C lags behind the current by 90 degrees



The waveform and equations for an RC series circuit can be drawn as below.

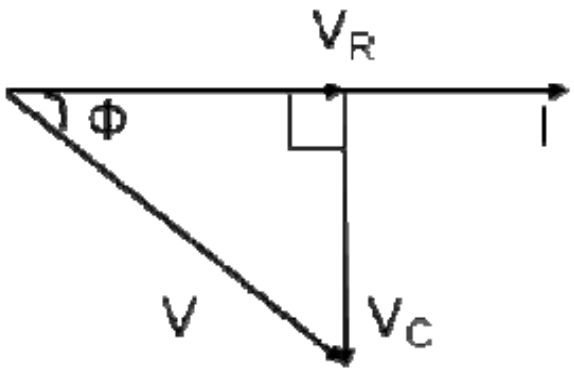


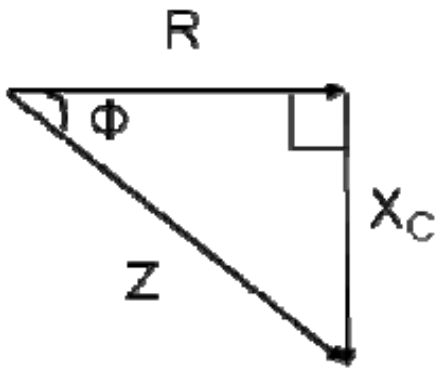
$$V = V_m \sin \omega t$$

$$I = I_m \sin(\omega t + \Phi)$$

Impedance Triangle

We can derive a triangle called the impedance triangle from the phasor diagram of an RC series circuit as shown





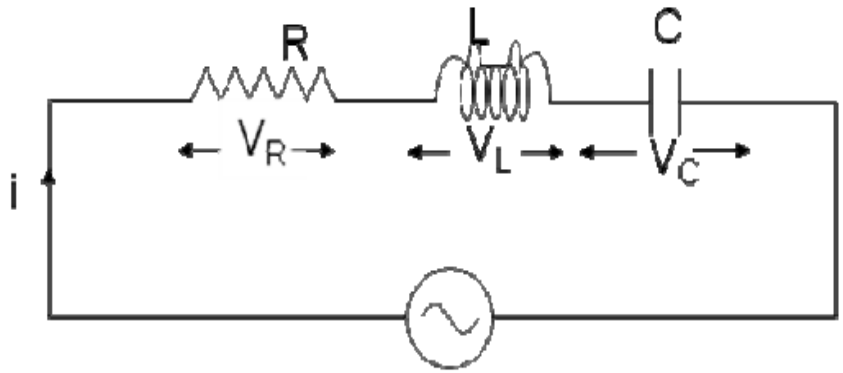
Phasor algebra for RC series circuit

$$V = V + j0 = V\angle 0^\circ$$

$$\overline{Z} = R - jX_C = Z\angle -\Phi$$

$$\overline{I} = \frac{\overline{V}}{\overline{Z}} = \frac{V}{Z}\angle +\Phi$$

R-L-C Series circuit



$$V = \sqrt{V_R^2 + (V_L - V_C)^2}$$

$$V = \sqrt{(IR)^2 + (IX_L - IX_C)^2}$$

$$V = I\sqrt{R^2 + (X_L - X_C)^2}$$

$$V = IZ$$

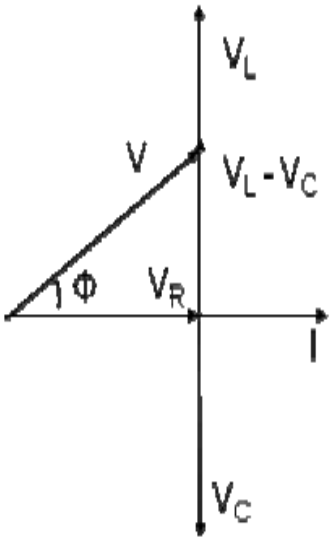
Where impedance $Z = \sqrt{R^2 + (X_L - X_C)^2}$

Phase angle

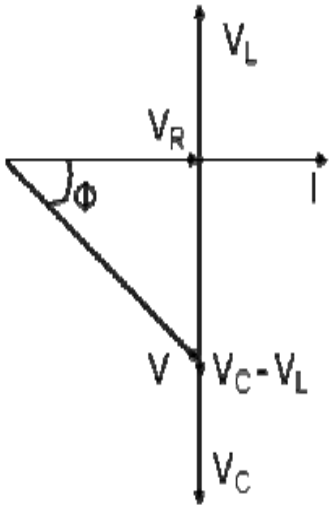
$$\Phi = \tan^{-1}\left(\frac{V_L - V_C}{V_R}\right)$$

$$\Phi = \tan^{-1}\left(\frac{IX_L - IX_C}{IR}\right)$$

$$\Phi = \tan^{-1}\left(\frac{X_L - X_C}{R}\right)$$



$V_L > V_C$



$V_L < V_C$

From the expression for phase angle, we can derive the following three cases

Case (i): When XL>XC

- The phase angle Φ is positive and the circuit is **inductive**. The circuit behaves like a series RL circuit.

Case (ii): When XL<XC

- The phase angle is negative and the circuit is **capacitive.** The circuit behaves like a series RC circuit.

Case (iii): When XL=XC

- The phase angle $\Phi = 0$ and the circuit is **purely resistive**. The circuit behaves like a pure resistive circuit.

Problem 29. The power taken by an inductive circuit when connected to a 120 V, 50 Hz supply is 400 W and the current is 8 A. Calculate (a) the resistance, (b) the impedance, (c) the reactance, (d) the power factor, and (e) the phase angle between voltage and current.

(a) Power $P = I^2R$ hence $\boldsymbol{R} = \dfrac{P}{I^2} = \dfrac{400}{8^2} = \mathbf{6.25\ \Omega}$.

(b) Impedance $\boldsymbol{Z} = \dfrac{V}{I} = \dfrac{120}{8} = \mathbf{15\ \Omega}$.

(c) Since $Z = \sqrt{R^2 + X_L^2}$, then
$X_L = \sqrt{Z^2 - R^2} = \sqrt{15^2 - 6.25^2} = \mathbf{13.64\ \Omega}$.

(d) $$\textbf{Power factor} = \frac{\text{true power}}{\text{apparent power}} = \frac{VI\cos\phi}{VI}$$
$$= \frac{400}{(120)(8)} = \mathbf{0.4167}$$

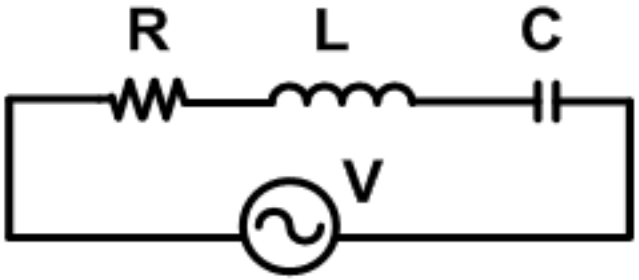
(e) p.f. $= \cos\phi = 0.4167$ hence
phase angle, $\phi = \cos^{-1} 0.4167 = \mathbf{65.37°\ lagging}$.

Problem(1):

- For the circuit shown In Fig.2. , for the following data is given Calculate **Z, I, Power, P.f., P & Q**

(A) R=10Ω, L=10mH, V=100V, if, f=25Hz

(B) R=15Ω, L=0.02H, V=210V,if, f=50Hz

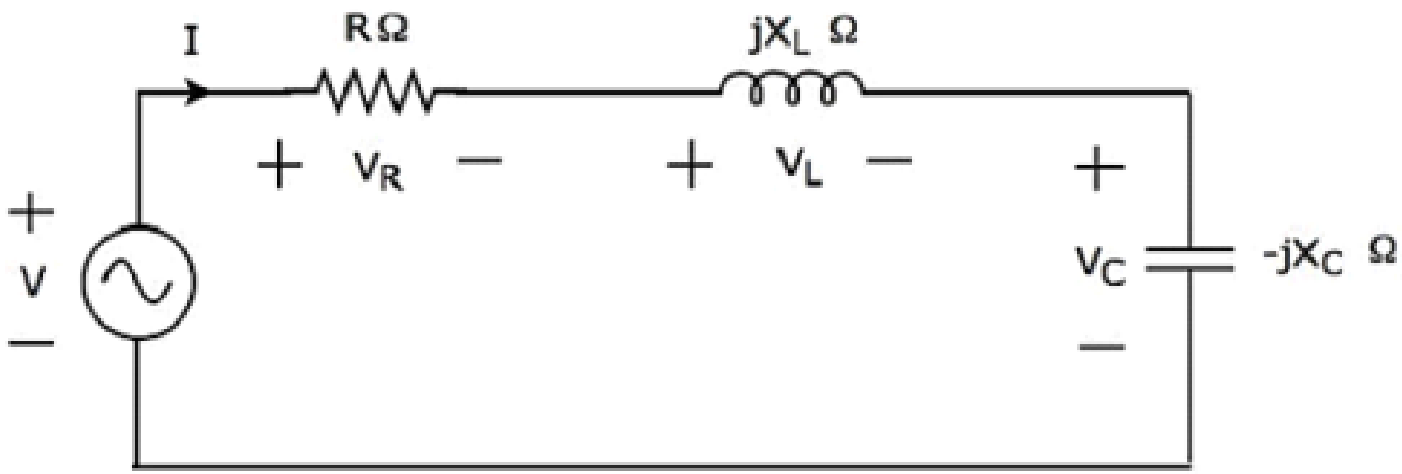


Resonance

- **Resonance** occurs in electric circuits due to the presence of energy storing elements like inductor and capacitor. It is the fundamental concept based on which, the radio and TV receivers are designed in such a way that they should be able to select only the desired station frequency.
- There are **two types** of resonances, namely series resonance and parallel resonance. These are classified based on the network elements that are connected in series or parallel.

Series Resonance Circuit Diagram

- If the resonance occurs in series RLC circuit, then it is called as **Series Resonance**. Consider the following **series RLC circuit**, which is represented in phasor domain.



Apply **KVL** around the loop.

$$V - V_R - V_L - V_C = 0$$

$$V - I_R - I(jX_L) - I(-jX_C) = 0$$

$$V = I_R + I(jX_L) + I(-jX_C)$$

$$V = I[R + j(X_L - X_C)]$$

The above equation is in the form of ***V = IZ***.
Therefore, the **impedance Z** of series RLC circuit will be
$Z = R + j(X_L - X_C)$

- The frequency at which resonance occurs is called as **resonant frequency f_r**.

  In series RLC circuit resonance occurs, when the imaginary term of impedance *Z* is zero, i.e., the value of XL−XC should be equal to zero.
- XL=XC

Substitute XL=2πfL and XC=1/2πfC in the above equation.

- 2πfL=1/2πfC
- $f^2=1(2\pi)^2LC$
- $f_r=1/(2\pi)\sqrt{LC}$

Voltage across Resistor

The voltage across resistor is

- $V_R = IR$

Substitute the value of ***I*** in the above equation.

- $V_R = (V/R)R$
- $V_R = V$
- Therefore, the **voltage across resistor** at resonance is $\boldsymbol{V_R = V}$.

Voltage across Inductor

The voltage across inductor is $V_L=I(jX_L)$

- Substitute the value of ***I*** in the above equation.
- $V_L=(V_R)/R\ (jX_L)$
- $V_L=j(X_L/R)V$
- VL=jQV

Therefore, the **voltage across inductor** at resonance is VL=jQV

So, the **magnitude** of voltage across inductor at resonance will be

- |VL|=QV

Where ***Q*** is the **Quality factor** and its value is equal to XL/R

Voltage across Capacitor

The voltage across capacitor is $V_C = I(-jX_C)$

Substitute the value of *I* in the above equation.

- $V_C = [V/R]\ (-jX_C)$
- $V_C = -j[X_C/R]V$
- $VC = -jQV$

Therefore, the **voltage across capacitor** at resonance is $V_C = -jQV$

So, the **magnitude** of voltage across capacitor at resonance will be $|V_C| = QV$

Where ***Q*** is the **Quality factor** and its value is equal to X_C/R

- **Bandwidth of a Series Resonance Circuit**

  The distance between these two points, i.e. ($f_H - f_L$) is called the **Bandwidth**, (BW) and is the range of frequencies over which at least half of the maximum power and current is provided as shown in fig.

  The sharpness of the peak is measured quantitatively and is called the **Quality factor, Q** of the circuit. $Q = f_r / BW$.